



Does conditioning leave a structural memory?

Large-area evidence for the dislocation picture

Yinon Ashkenazy

Racah Institute of Physics, The Hebrew University of Jerusalem

I. Popov (HUJI) · W. L. Millar, V. M. Bjelland, W. Wuensch (CERN)

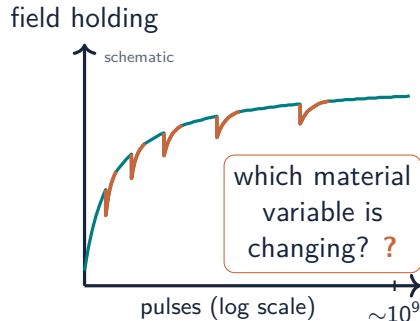
Montreux, Switzerland · Monday 5 October 2026



Conditioning works. What does it change?



- ▶ Conditioning is how every high-gradient device reaches its usable field.^a
- ▶ Breakdown rate has an extreme field dependence, $\sim E^{30}$.^a
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.^{a,b}
- ▶ The field, more than the arcs, drives conditioning.^{a,b}



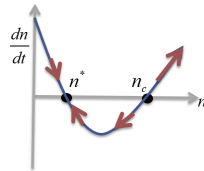
Something in the metal stores the improvement. We do not know what.^a

Sources: ^aWuensch et al., Rev. Mod. Phys. 98, 025004 (2026); ^bDegiovanni et al., PRAB 19, 032001 (2016).

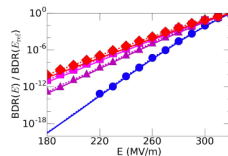
Our model: breakdown as collective dislocation dynamics



- ▶ Field holding rises FCC→BCC→HCP^d — matching the order of barriers to dislocation motion.
- ▶ The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.^{a,b}
- ▶ The same model, fitted to the $\sim E^{30}$ rate and its temperature dependence,^d then reproduced the histogram of dark-current spikes.^c



mobile-dislocation runaway

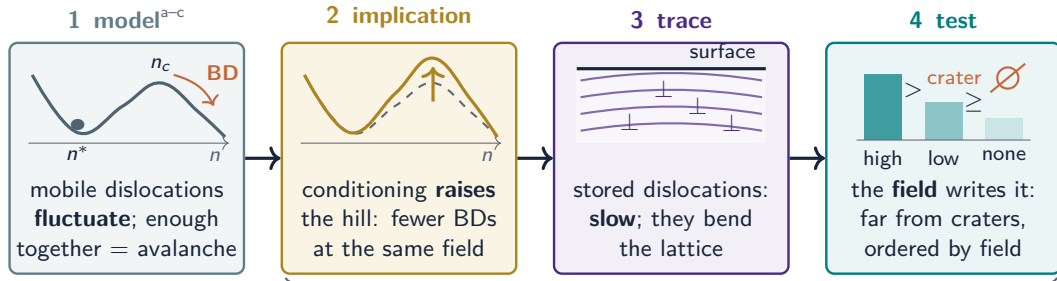


breakdown rate vs field

Central prediction: conditioning *evolves the near-surface dislocation structure*.^d

Sources: ^aEngelberg et al., PRL 120, 124801 (2018); ^bPRAB 22, 083501 (2019); ^cPRAB 23, 123501 (2020); ^dWuensch et al., RMP 98, 025004 (2026).

What the model says the microscope should see



inference from the model: no equation for conditioning yet

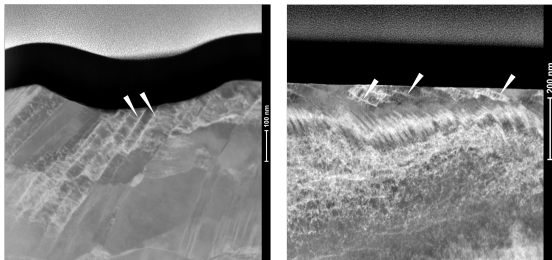
- ▶ In the model, a $\sim 10^9$ -pulse memory must sit in stored, EBSD-visible dislocations.
- ▶ The field writes it: look far from craters; expect the order to follow the local field.

predicts where, and in which order

does not predict size, depth, or sign

Sources: ^aEngelberg et al., PRL 120, 124801 (2018); ^bPRAB 22, 083501 (2019); ^cPRAB 23, 123501 (2020); Wuensch et al., RMP 98, 025004 (2026).

Identified using STEM at specific locations



Cross-sectional STEM of conditioned hard (as-machined)

Cu: dislocation-poor zone in the top ~ 200 nm.^a

STEM: HUJI, I. Popov

- ▶ **Direct** imaging of an altered near-surface dislocation structure in conditioned regions.^a
- ▶ Fewer dislocation walls at 300 K and 30 K; denuded layer clearest at 30 K.^a
- ▶ Fits the clue that soft Cu conditions far more slowly than hard Cu: the starting population matters.^b
- ▶ FIB preparation plants its own dislocations; trust only large-scale structure.
- ▶ One cross-section cannot speak for the whole electrode.

Seen at specific locations. The large-area view was missing.

Sources: ^aJacewicz et al., JAP 137, 193302 (2025); ^bKorsbäck et al., PRAB 23, 033102 (2020).

A sub-yield stress, repeated a billion times



Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: 0.028 MPa^c

$\sim 10^3 \times$ below the yield of annealed Cu^c



- ▶ Fatigue at 10^8 – 10^{10} cycles rearranges dislocations irreversibly, well below yield.^{a,b}
- ▶ Here: $\sim 10^9$ conditioning pulses.^c
- ▶ Pulsed DC means **no RF pulsed heating** and no bulk currents. Maxwell stress is the dominant cyclic load.^c
- ▶ MDDF needs only the stress that moves **mobile dislocations in a frustrated crystal**: far below yield.

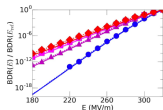
**The model predicts a large-area signature.
Is it there?**

Sources: ^aMughrabi, MMTB 40, 431 (2009); ^bStanzl-Tschegg et al., Int. J. Fatigue 29, 2050 (2007); ^cAshkenazy et al., arXiv:2606.19192 (2026).

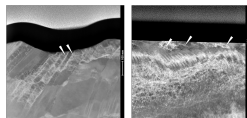
Last year we asked for more forensics. Here it is.



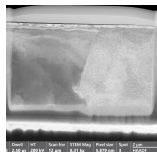
MeVArc 2025



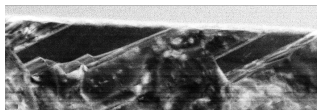
MDDF model



local STEM denuded zone^a



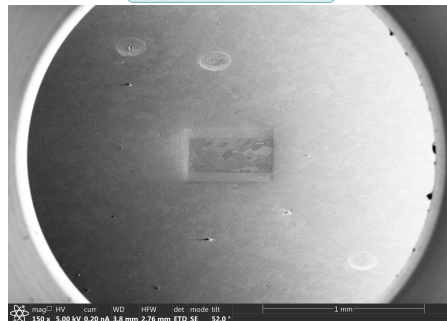
FIB lamella,
HAADF overview



cross-section STEM
at the surface

STEM: HUJI, I. Popov

MeVArc 2026

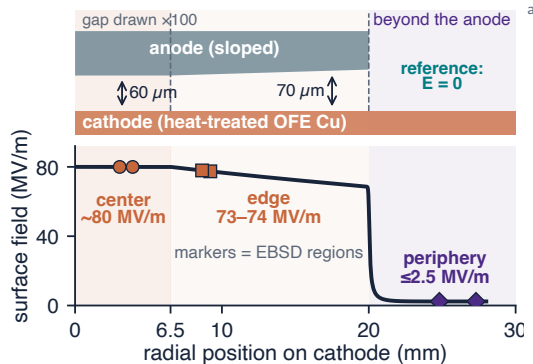


Large-area EBSD across one cathode

Closed with: other materials? more samples?

Source: ^aJacewicz et al., JAP 137, 193302 (2025).

One cathode turns field exposure into a controlled coordinate



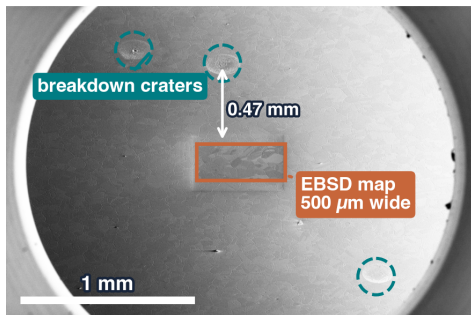
- ▶ Heat-treated OFE Cu, pulsed DC
- ▶ 1 μs pulses at 1 kHz, $\sim 10^9$ pulses
- ▶ Peak field ~ 80 MV/m at center
- ▶ Nine EBSD regions: center, edge, periphery, and external reference

● high field (center, edge) ● periphery (≤ 2.5 MV/m) ● unexposed reference

The field gradient gives us internal controls on a single specimen.

Sources: ^ageometry: Bjelland et al., arXiv:2606.21259 (2026); Ashkenazy et al., arXiv:2606.19192 (2026), $E(r)$: Fig. 3.

EBSD maps plastic activity far from breakdown craters



- ▶ Large area where TEM could only sample a point
- ▶ 500 μm -wide maps; 3 μm step (lateral only)^a; information depth ~ 40 nm at 15 kV^{b,c}
- ▶ $\sim 24,000$ indexed points per ROI^a
- ▶ **LAM** = local average misorientation: lattice tilt between a point and its neighbors (KAM, LOS: cross-checks)^a
- ▶ Misorientation is a proxy for GND density^d

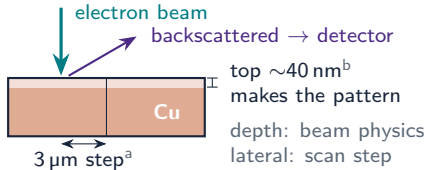
Target: the cumulative response to the field, measured far from craters.^a

Sources: ^aAshkenazy et al., arXiv:2606.19192 (2026); ^b ~ 40 nm at 15 kV: CASINO Monte Carlo, Drouin et al., Scanning 29, 92 (2007); ^cmeasured 38–72 nm over 5–30 kV, Chen et al., Ultramicroscopy 111, 1488 (2011); ^dPantleon, Scr. Mater. 58, 994 (2008).

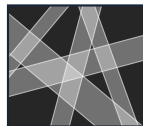
What one EBSD pixel measures



1 15 kV electron beam, point by point



2 Pattern → lattice orientation



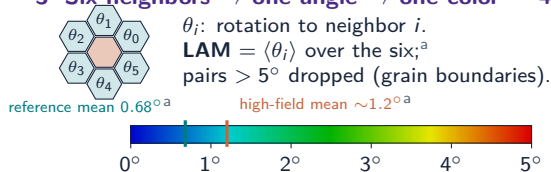
Kikuchi pattern



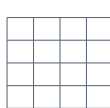
one lattice orientation for this point

~24,000 points^a per region, unfiltered

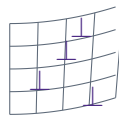
3 Six neighbors → one angle → one color



4 Why a rotation angle counts dislocations



flat: $\theta = 0$



curved: $\theta > 0$

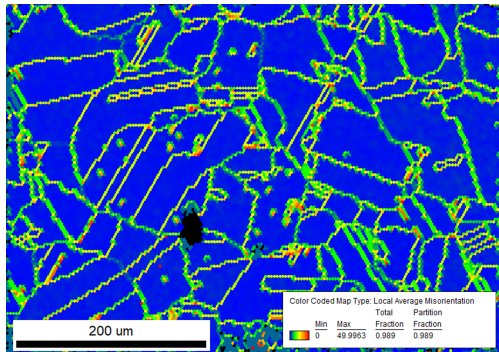
a dislocation proxy over 500 μm, not a dislocation image

Bending a crystal by an angle over a distance needs **geometrically necessary dislocations**, so **LAM** is a proxy for GND density.^c

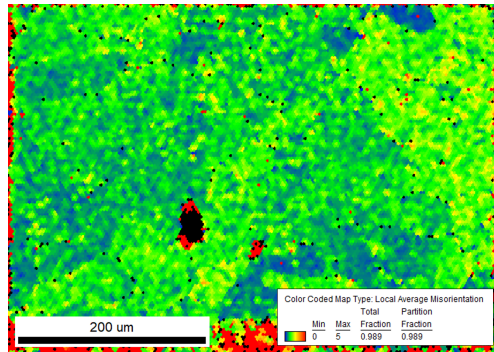
Each 3 μm point yields one angle, read from the top ~40 nm of copper.^b

Sources: ^aAshkenazy et al., arXiv:2606.19192 (2026); ^binformation depth: Chen et al., Ultramicroscopy 111, 1488 (2011); ^cPantleon, Scr. Mater. 58, 994 (2008).

How to read a LAM map: wide-angle vs low-angle



0–50°: grain boundaries dominate

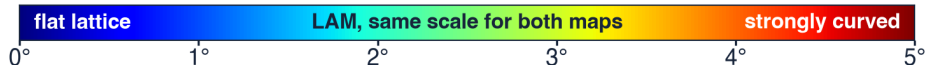
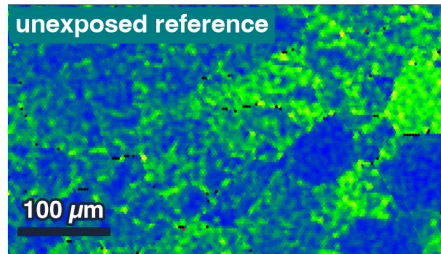
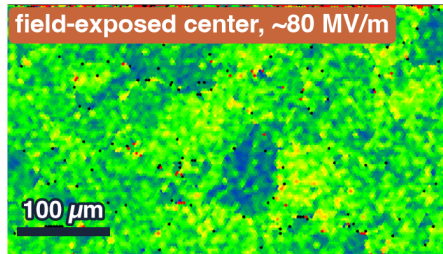


0–5° scale: intragrain curvature

Rescaling to 0–5° exposes the intragrain misorientation we care about.

Source: Ashkenazy et al., arXiv:2606.19192 (2026). Same area, two color scales.

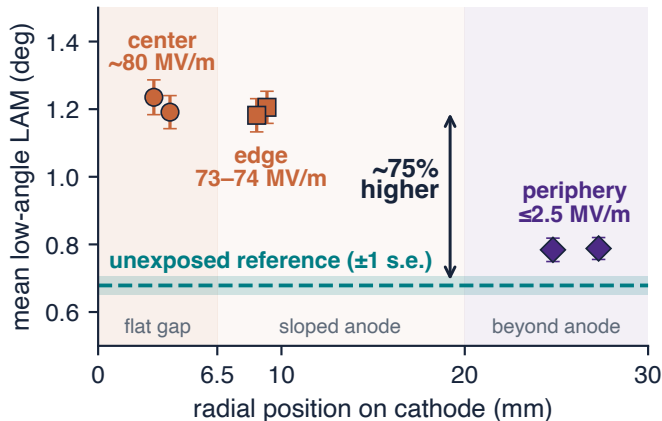
High-field copper contains more intragrain curvature



Same scale, same magnification: field-exposed grains carry more curvature.

Source: Ashkenazy et al., arXiv:2606.19192 (2026). Both maps cropped to the same $485 \times 279 \mu\text{m}$ field; LAM $0-5^\circ$.

The signal separates into three exposure tiers

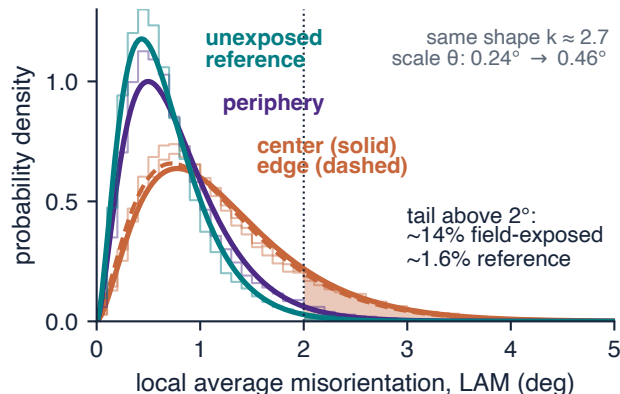


- ▶ Center and edge: $\sim 1.2^\circ$
- ▶ External reference: $\sim 0.68^\circ$
- ▶ High field is $\sim 75\%$ higher
- ▶ Grain size does not follow this hierarchy

The ordering follows field history; grain size does not track it.

Source: Ashkenazy et al., arXiv:2606.19192 (2026). Error bars: grain-corrected standard errors; band: unexposed reference ± 1 s.e.

Field exposure builds a high-misorientation tail



LAM tail above 2°

~ 0.14 field-exposed
 ~ 0.016 reference

$\sim 8\times$ **increase**

Gamma shape $k \approx 2.7$ fixed; the scale roughly doubles.

The whole distribution stretches; a uniform offset cannot do that.

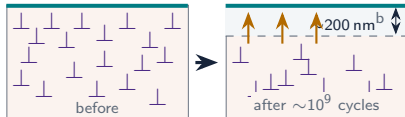
Source: Ashkenazy et al., arXiv:2606.19192 (2026). One ROI per tier (histograms faded, gamma fits solid, edge dashed); shaded: tail above 2° .

The large-area test supports the model's prediction



Hard, as-machined Cu

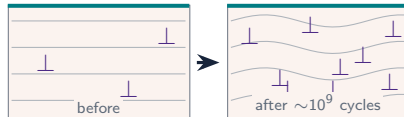
dense starting population (2025 STEM cathodes)



dislocations escape to the surface or annihilate $\Rightarrow$ **depleted layer (STEM)**^b

Heat-treated OFE Cu

sparse starting population (this cathode)



one load
 $\sigma_M = \frac{1}{2} \epsilon_0 E^2$
 $\sim 10^9$ cycles
 10^{-3} of yield

the manuscript's interpretation^a

same load generates and rearranges dislocations $\Rightarrow$ **lattice curvature (EBSD, LAM $\sim 75\%$ higher)**^a

- To our knowledge, the first **large-area** observation that conditioned regions carry a different near-surface dislocation structure, ordered as MDDF predicts.^a
- The measured order follows the simulated local field:
center $\approx$ edge $>$ periphery $>$ reference.^a

Before, dislocations were plausible.

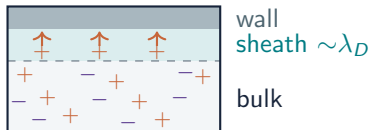
Now a dislocation-sensitive state tracks field exposure.

Sources: ^aAshkenazy et al., arXiv:2606.19192 (2026), discussion; ^bJacewicz et al., JAP 137, 193302 (2025).

Elastic screening: a sheath of dislocations at the surface



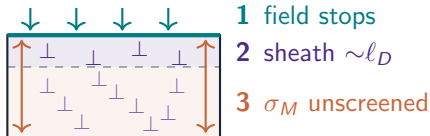
Plasma at an absorbing wall schematic



1 $\lesssim 0.1 \text{ nm}$
field penetration
(Thomas–Fermi)

2 $\ell_D \sim 25 \text{ nm}^a$
screening of weak internal
fluctuations;^{b,c} only this layer
talks to the surface

Copper under the Maxwell traction schematic



3 $\sim \text{mm}$
applied load, not screened
(dislocation stress is
self-equilibrated)^d

In this picture, weak screened internal stresses set the structure; the strong unscreened load drives the mobile dislocations through it.

Sources: ^aGroma et al., PRL 96, 165503 (2006); ^bLemaître et al., PRE 104, 024904 (2021); ^cLivne et al., PRE 107, 055004 (2023); ^dAshkenazy et al., arXiv:2606.19192 (2026).

Copper shows the effect, not universality



What is shown

- ▶ Large-area structural difference
- ▶ Three metrics agree (LAM, KAM, LOS)
- ▶ Internal and external controls
- ▶ Supports the MDDF prediction

What remains open

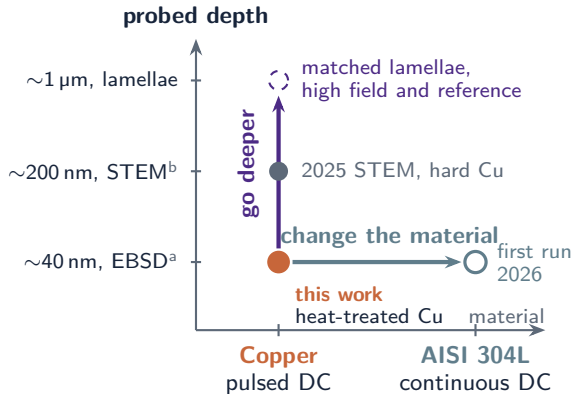
- ▶ One conditioned cathode, one experiment
- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

Limits of this dataset: field level and breakdown density vary together by construction ($\sim 24, 12, 5$ breakdowns per cm^2 , center outward); LAM values are relative, compared only between regions mapped under the same step and protocol.

A stronger test changes both the material and the loading history.

Source: Ashkenazy et al., arXiv:2606.19192 (2026), discussion and limitations.

Next: go deeper, and change the material



- ▶ Cross-sections from a mapped high-field region and from the reference, through the top $\sim 1\ \mu\text{m}$: do the EBSD layer ($\sim 40\ \text{nm}$) and the STEM denuded zone ($\sim 200\ \text{nm}$) belong to one depth profile — is the denuded zone the sheath ($\approx 8\ell_D$)?
- ▶ Artifact controls from the 2025 “red herrings”
- ▶ Stainless steel under continuous DC, with Consorzio RFX — next slide

Depth continues our STEM line; material tests the mechanism itself.

Sources: ^aCASINO Monte Carlo, Drouin et al., Scanning 29, 92 (2007); ^bJacewicz et al., JAP 137, 193302 (2025).

Steel: different material, different structural evolution



What differs from copper

- ▶ Lower stacking-fault energy: planar slip, different dislocation mobility
- ▶ Continuous DC: a steady traction of ~ 16 kPa, not $\sim 10^9$ pulses
- ▶ Metastable austenite is a built-in strain gauge: ε and α' martensite mark plastic strain above a few 0.1 %^a

The dislocation picture expects a different near-surface evolution. The test: does any field-correlated change appear at all?



Consorzio RFX — De Lorenzi, Pilan, Spada

First run with Consorzio RFX

- ▶ AISI 304L at the RFX HVPTF: continuous DC, 90 h and 73 h at 59–61 MV/m; zero breakdowns, stopped at emission switch-on
- ▶ 26 EBSD maps (HUJI, I. Popov): the exposed apex lies inside or below the unexposed range
- ▶ References exist (unexposed electrodes, side regions) but no same-area before-and-after; with 2–4 maps per group a field effect cannot yet be resolved

RFX HV program: N. Pilan, next talk (12:30).

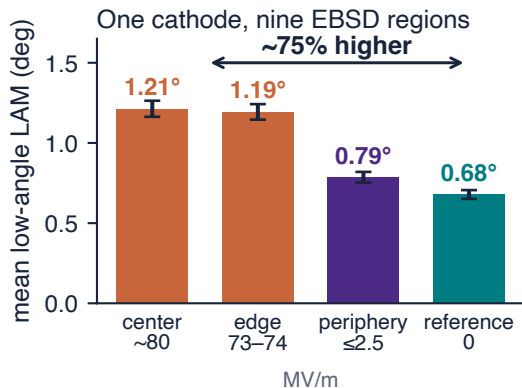
Copper supports the memory; steel asks whether it is universal.

Source: ^aBayerlein et al., MSEA 114, L11 (1989).

Conditioning appears to leave a subsurface memory



1. Our model predicts conditioning evolves the near-surface dislocation structure.
2. 2025 STEM saw it locally; **large-area EBSD now supports it** ($\sim 75\%$ higher misorientation).
3. Depth and stainless steel test whether the memory survives a change of scale, alloy, and loading.



Last year: “other materials? more samples?” — here is the large-area answer.

Copper result: Ashkenazy et al., arXiv:2606.19192 (2026).

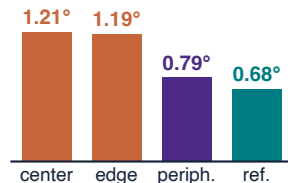
Conclusions, and an invitation



1. Conditioning leaves a structural memory in the near-surface dislocation population — supported on one cathode, not yet replicated.^a
2. Our picture: it lives within a screening length of the surface — the layer EBSD sees, and the layer that sets emission.
3. What would settle it: the same area mapped before and after; a depth profile; a second material (steel, with Consorzio RFX).

Talk to me: new materials, other electrode configurations, cross-facility comparisons, same-area before-and-after mapping — let us test this together.

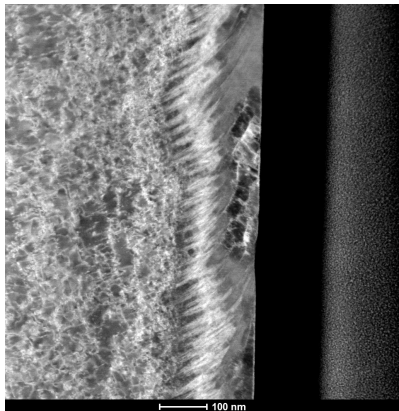
mean LAM, one cathode



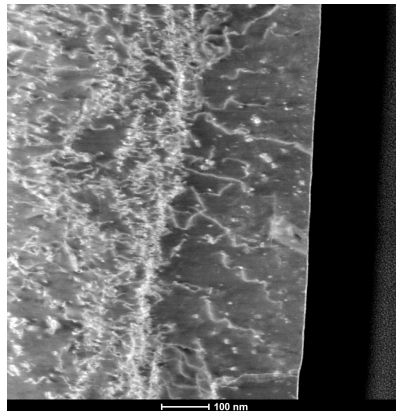
arxiv.org/pdf/2606.19192

^aAshkenazy et al., arXiv:2606.19192

Backup: field-exposed vs reference STEM (2025)



Field-exposed, 300 K: fewer walls



Reference, 300 K: walls to surface

STEM: HUJI, I. Popov.

Source: Jacewicz et al., J. Appl. Phys. 137, 193302 (2025), Figs. 8–10.